Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **object-oriented, data-intensive applications** across C#, Java, Python, SQL, and Linux, translating stakeholder requirements into reliable integrations, production releases, and measurable performance improvements.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected asynchronous FastAPI services and Azure SQL workflows, processing 1K+ concurrent jobs at 3x throughput through **object-oriented design, indexed queries, fault isolation, monitoring, and responsive production support**.
- Translated stakeholder workflows into resilient platform features for 500+ users, reducing dashboard latency 71% through **requirements gathering, SQL optimization, deployment ownership, systematic troubleshooting, and iterative user adoption**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Delivered React and Flask applications supporting 150+ instructors and 10K+ daily requests through **cross-functional requirements analysis, REST API development, code reviews, automated testing, debugging, and production releases**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **performance profiling and failure recovery**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Developed cross-platform integrations using C#, Java, SQL, and object-oriented patterns, improving API response times 30% while reducing technical debt 20% across distributed, business-critical data-sharing services and workflows.
- Automated 20+ unit and integration suites across Windows and Linux using PyTest, Mockito, Jenkins, and virtual machines, strengthening **CI/CD validation, regression detection, and dependable production release support**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built C# ASP.NET Core services and React interfaces for resources, billing, permissions, and notifications, synchronizing 2M+ records through **SQL-backed APIs, WebSockets, authentication, and modular enterprise integration workflows**.
- Automated Docker and Jenkins deployments, cutting release time 50% while using Sentry, Azure Monitor, and Kibana for **real-time issue diagnosis, root-cause analysis, incident response, and production reliability**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Gathered client requirements and presented technical designs for 12+ integrations, mapping booking, payment, and status workflows while **bridging business and engineering communication** and reducing design errors 25%.
- Built Flask and Firebase APIs supporting 50K+ records with sub-200-millisecond responses, delivering **live demonstrations, urgent bug fixes, integration guidance, and responsive stakeholder support** through implementation and production release.

Skills

Languages & Frameworks: C#, Java, C++, Python, TypeScript, JavaScript, SQL, ASP.NET Core, React, Flask, FastAPI, Entity Framework
Trading & Integration: Trading Pipelines, Market Data, Order Management, REST APIs, WebSockets, JSON, Authentication, System Integration, Real-Time Support
Data & Systems: PostgreSQL, Azure SQL, SQL Server, MySQL, Redis, Data Modeling, Concurrency, Multithreading, Linux, Windows, Distributed Systems
Cloud & Delivery: Azure, AWS, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Monitoring, Logging, Sentry, Azure Monitor, Kibana
Engineering: Object-Oriented Programming, Design Patterns, Data Structures, Algorithms, Unit Testing, Integration Testing, Debugging, Performance Profiling, Agile, Code Reviews, Requirements Gathering, ChatGPT Codex

Projects

TradeBridge Integration Platform | *C#, .NET 8, SQL Server, FIX, WebSockets, Redis*

- Designed a **real-time trading integration platform** connecting market-data feeds, order workflows, and risk checks through asynchronous C# services, idempotent message handling, SQL persistence, and operator-facing reconciliation tools.

LiveOps Trading Console | *C#, React, Python, PostgreSQL, Linux, OpenTelemetry*

- Built a **live trading support console** correlating orders, logs, positions, and service health, with alert triage, replayable diagnostics, audit trails, and guided remediation for rapid production bug resolution.

CPU Scheduling Simulator | *C++, React, JavaScript, Priority Queues, Multithreading* | Live

- Implemented performance-sensitive C++ algorithms for FCFS, SJF, SRTF, Round Robin, priority, and multilevel queues, validating correctness and visualizing runtime behavior across **interactive scheduling, resource allocation, and concurrency simulations**.